

The geophysics of the Olympic Dam discovery

H. Rutter & D. J. Esdale

To cite this article: H. Rutter & D. J. Esdale (1985) The geophysics of the Olympic Dam discovery, Exploration Geophysics, 16:2-3, 273-276, DOI: [10.1071/EG985273](https://doi.org/10.1071/EG985273)

To link to this article: <https://doi.org/10.1071/EG985273>



Published online: 06 Dec 2018.



Submit your article to this journal [↗](#)



Article views: 14



View related articles [↗](#)



Citing articles: 2 View citing articles [↗](#)

Peter J. Clark received his BA in physics and electronics from Macquarie University, Australia in 1979 and his BSc from New England University, 1982. He was a visiting lecturer in geophysics at the University of Papua New Guinea in 1983. He has been an experimental scientist at the Division of Mineral Physics of the CSIRO, Sydney, since 1984. His research interests include magnetic gradiometry, EM inversion and alternative geophysical methods of petroliferous exploration.

P. Clarke, CSIRO Division of Mineral Physics, PO Box 136, North Ryde, NSW 2113.



Bob Smith is a graduate of the University of Melbourne, where he completed a BSc(Hons) in physics in 1960. Since then he has worked as a geophysicist with the Bureau of Mineral Resources in Melbourne and Canberra, Chief Geophysicist and Manager (Geophysical Services) with McPhar Geophysics Pty Ltd in Adelaide, a private consultant and Chief Geophysicist for CRA Exploration Pty Ltd, based in Adelaide.

R. Smith, CRA Exploration Pty Ltd, PO Box 254, Norwood, SA 5067.

The geophysics of the Olympic Dam discovery

H. Rutter and D. J. Esdale

The central southern part of South Australia was one of a number of areas considered favourable for economic deposits of copper mineralization by WMC geologists in 1974 (Fig. 1). This area was far too large to be explored in detail. The process of ground elimination and target selections required the use of all the available geological and geophysical information.

The geological model was that of a sediment hosted copper deposit where the metal source was considered to be a thick pile of continental tholeiite basalt lavas; mobility was to be provided by hydrothermal fluids active during metamorphic alteration processes. This was translated into a geophysical model which consisted of a magnetic anomaly caused by the tholeiitic basalts and a gravity anomaly related to a horst block within the main basaltic mass. The object was to identify this situation within the geological province using the regional gravity and magnetic data.

Samples of Roopena Volcanics (of Willouran age) considered to be similar to the source rock, were obtained from exposures at Roopena Station and Depot Creek both close to Port Augusta. Measurements of magnetic susceptibility and density were made and compared to the same parameters measured on samples of Marinoan rocks (Whyalla Sandstone, Tregolana Shale, Corraberra Sandstone, Arcoona Quartzite, Tent Hill Formation). The susceptibility contrast was variable and related to the degree of alteration of the basalt samples, but a density contrast of 0.3 gms/cc was established with a greater degree of credibility.

The regional magnetics were first assessed with the object of locating a basaltic pile of sufficient thickness, lateral extent,

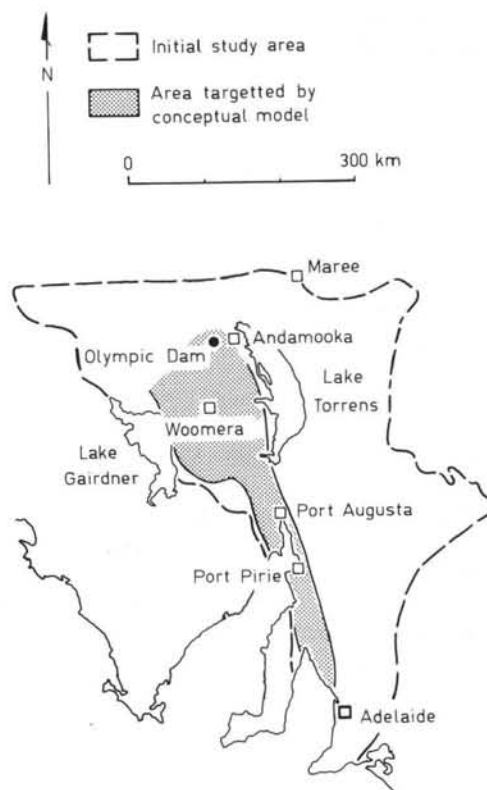


Fig 1 Location of Olympic Dam copper deposit and initial study area.

and nearness to surface to fulfil the conditions of the geological model while still being amenable to exploration.

It soon became apparent that the magnetic anomalies on the Torrens and Andamooka 1:250 000 sheets were of prime interest, although far more distant areas were also considered. The magnetic complex on the Andamooka sheet satisfied many of the criteria for extensive, thick, and faulted basalts, and was subjected to a first stage magnetic interpretation. Six anomalies were selected as being caused by a source relatively close to the surface; of these, five were on the Andamooka sheet and became known as Bills Lookout, Olympic Dam, Appendicitis Dam, Andamooka Island and Lake Torrens. All five were also associated with a positive gravity anomaly.

The original gravity map published by SADM consisted of 2 mg/l contours compiled from stations spaced on a 6 km grid. This data was carefully recontoured at a 1.0 mg/l interval in the area of each anomaly and from the new contours a profile was taken for interpretation. Simple geometric models were used in conjunction with a density contrast of 0.3 mg/l to simulate uplifted blocks of basaltic material in a sedimentary environment. The errors that could result from an interpretation based on such widely spaced data relative to the size of the source were fully realised; it was expected that the true anomaly would be smoothed and the source interpreted to be too deep.

The estimated depth to the source of the gravity anomaly at Andamooka Island was 1700 m; Lake Torrens 2700 m; Bills Lookout 2200 m; Appendicitis Dam 4800 m; and Olympic Dam 1150 m. As a result the Olympic Dam anomaly was one of those selected for further investigation; and it soon became apparent that by adjusting the physical dimensions of the body but maintaining the density contrast the source could be as shallow as 850 m.

An interpretation of the total magnetic intensity, again using simple models, placed the depth to the magnetic source at about 2000 m. Relating the magnetic and gravity interpretations to the original model was becoming difficult because of the inconsistent depths; but it was considered still possible because of the regional nature of the data and the possible interpretation errors that could have been incurred.

Olympic Dam was also favoured because of accessibility because the Lake Torrens and Andamooka Island anomalies were both beneath Lake sediments of unknown thickness and Bills Lookout anomaly was in the Andamooka Opal Field, an area restricted to opal mining and inaccessible to general exploration.

Ground inspection followed, but there was little to see on the ground other than sand dunes, some Cretaceous outcrop, and in the vicinity of Olympic Dam, outcropping Cambrian limestones. A ground magnetometer traverse was run from a point 2 km SSE of Olympic Dam, through Olympic Dam to Coronation Dam and on to the Vermin Proof Fence (Fig. 2). The profile form is identical to the airborne profile and other than confirming the ground location of the anomaly, no further information concerning the source was achieved. It is relevant to note that a small speck of malachite on a sandstone rock was found a short distance north of Olympic Dam while on this magnetic traverse. It had little to do with the subsequent mineral discovery but it added encouragement at the time. Confirmatory geophysical evidence was sought from seismic refraction and reflection surveys. Using equipment and personnel from SADM three refraction seismic traverses were

run firstly along the road from Axehead Dam to Phillips Ridge Hut; secondly along the N-S fence towards Olympic Dam from Phillips Ridge Hut and thirdly west from this fence to Olympic Dam and slightly beyond (Fig. 3). Reflection seismic data was collected only at Olympic Dam, but the results were the most significant in that they identified a strong reflector at about 1000 ft (305 m).

The geophysical signatures recorded at Olympic Dam were those expected to result from the conceptual geological model. The magnetics reflected extensive thicknesses of tholeiitic basalts in the area with shallower parts at Olympic Dam; the gravity confirmed the presence of denser rock; and the seismic indicated that optimistically this might be at a depth as shallow as 300 m. RD 1 was sited 200 m west of Olympic Dam, 10 June 1975 and drilling began four days later. It was a stratigraphic hole to determine the lithologies and check for the presence of basalts in the context of the initial conceptual geological model. RD 2 was sited at Appendicitis Dam (now named Acropolis) 18 km southeast, using only the regional gravity and airborne magnetics. RD 1 was completed to 411 m, intersecting the following succession: Andamooka Limestone 0–37.9; Arcoona Quartzite 37.9–197.5; Corraberra Sandstone 197.5–209.9; Tregolana Shale 209.0–334.9; Medium to coarse grained hematitic sandstone 334.9–335.7; Coarse grained altered hematitic amygdoloidal basic volcs 335.7–411.0.

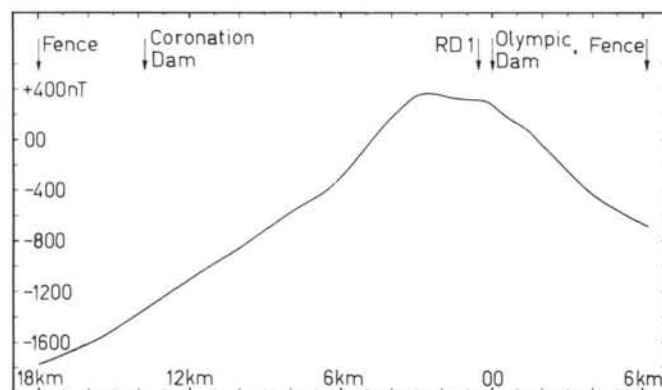


Fig 2 Ground magnetic traverse Coronation Dam-Olympic Dam relative values.

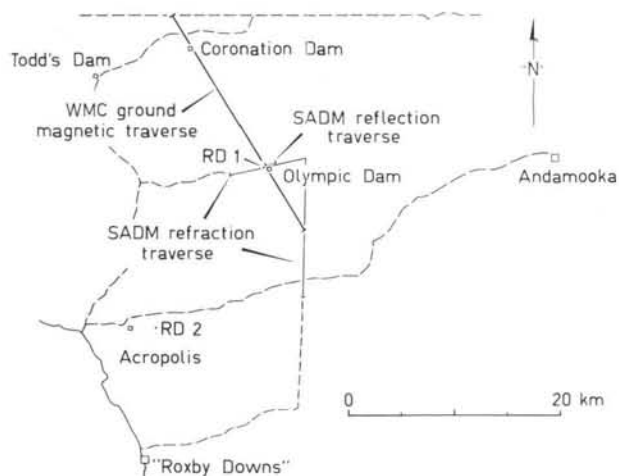


Fig 3 Geophysical ground surveys.

Subsequent assays gave 38 m of 1.05% Cu in the lowest unit which was not magnetic but had a high density which went far in explaining the gravity anomaly. The hematitic brecciated rock could be the cause of the gravity but not the magnetic anomaly. The upper surface could be the seismic reflection detected from the surface, but so could the top of the coarse sandstone, only a metre higher up the hole.

It was decided to run a seismic refraction traverse from RD 1 to RD 2 to determine which horizon was the refractor at RD 1, follow it to RD 2 and relate the geological intersections of both drillholes. As a result of this survey the refractor was found to be the Tregolana shale and Tent Hill Fm interface.

Gravity and magnetic measurements were taken at 90 m intervals along this line and also along a line perpendicular at both RD 1 and RD 2. The contoured gravity data at Olympic Dam indicated a peak NE of the drillhole.

An Elmac logger was used to record the natural gamma radiation in RD1, with rather spectacular results in the lowest unit. The appropriate sections of the core were examined using a spectrometer with the surprising result that the cause was not just potassium but predominantly uranium.

Considerable interest was generated in the Olympic Dam prospect with its unusual rock type and anomalous copper and uranium to the extent that a seismic reflection crew, on its way to the Pedirka Basin, was diverted for a brief test traverse across Olympic Dam. It is doubtful that the results were particularly meaningful because of the noise generated in the cavernous Cambrian limestone but it is an indication of the interest being placed in the prospect at that time. Drilling continued but the results were not spectacular.

A detailed gravity grid was surveyed over the area with lines 500 m apart and stations at 100 m intervals (Fig. 4). The data placed the centre of the gravity anomaly 1.3 km north of Olympic Dam and outlined a feature which was approximately 2.5 km in diameter.

A conventional IP line was run across the centre of the feature: using a dipole-dipole array with 100 m dipoles to $n=8$: both apparent resistivity and percentage frequency effect data were gathered. There was no obvious relationship between either parameter and subsurface mineralization.

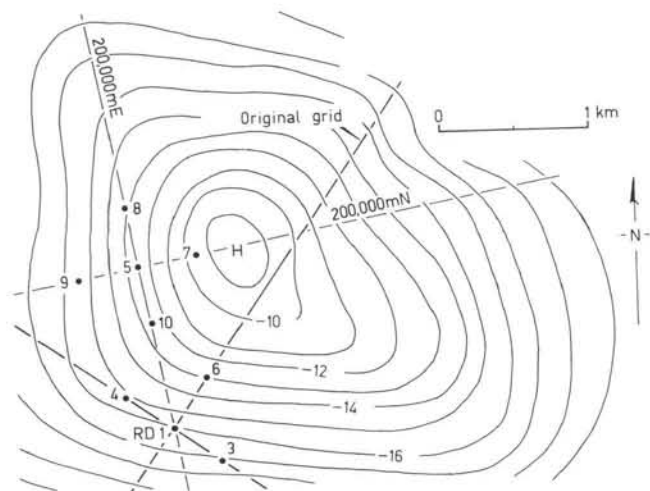


Fig 4 Detailed gravity survey and drillhole location. Contours in milligals.

The results of RD 10 became available in November 1976 (17 months after RD 1) which included an intersection of 168 m at 2.13% Cu and later in that same month the results of all drillholes RD 1-RD 10 were announced at the WMC Annual General Meeting.

The original conceptual model had not been found. The stratigraphic hole RD 1 had located a hematitic brecciated granitic rock and not tholeiitic basalts: if the latter had been intersected, exploration would have moved to locate the copper mineralization which was hoped would surround this centre in the draped sediments.

The deposit is quite different. The gravity anomaly is quite clearly caused by the prodigious amounts of hematite. The magnetic anomaly remained unexplained up to the stage of drilling RD 10, because the interpreted depth of the source, 1800 m, had not been reached.

The uranium was in equilibrium and apart from using the gamma logs to define uranium rich sections, the logs were also used to estimate uranium content prior to a formal geochemical analysis. The regional gravity and magnetic data provided the basis for the original study which located the site of RD1. Even though the data is widespread and interpretations of the gravity data overestimated depth, the relative depths were correct; Olympic Dam was the shallowest feature. The regional gravity was enhanced by reprocessing and contouring at a 1.0 mg interval (Fig. 5).

Late in 1976 with the RD 10 intersection, the realization of the size of the deposit, and the effort required for evaluation became apparent. A programme of drilling was established and various studies in many facets of the Earth Sciences were proposed. The results of these studies, in particular those of geophysical significance, will, no doubt, become available as time progresses.

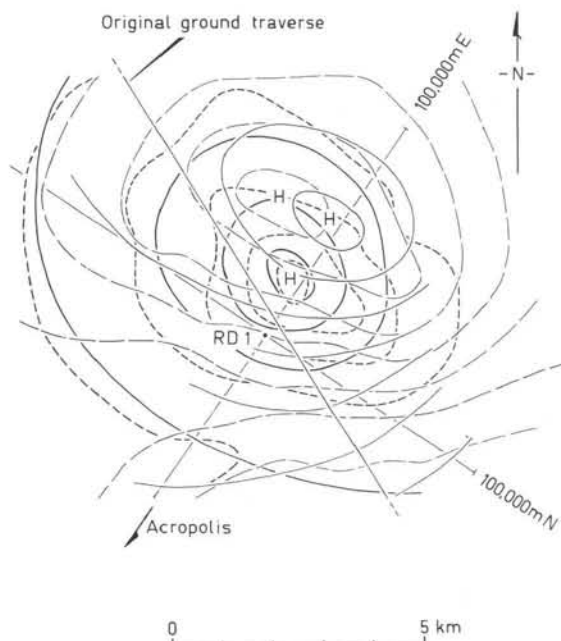


Fig 5 Geophysical summary. Magnetic contours: — original; - - detail. Gravity contours: — original; - - detail.



Hugh Rutter graduated in geology from Durham University in 1963, subsequently receiving an MSc in geophysics from London University and a DIC from the Royal School of Mines. He worked for 3 years with the Institute of Geological Sciences in London before emigrating to Australia in 1967. For 18 months he was employed as a geophysicist by the Geological Survey of WA, and then joined Western Mining Corporation in Kalgoorlie. When WMC expanded to the Eastern States he was transferred as Senior Geophysicist to Melbourne. During the period with WMC he was actively involved in a number of major mineral discoveries. In 1977 he joined the Broken Hill Proprietary Co. Ltd as Chief Geophysicist in the Minerals Division. He is now Consulting Geophysicist and Director of Geophysical Exploration Consultants Pty Ltd, and Honorary Lecturer in Geophysics at Monash University. He is a member of SEG, ASEG, EAEG, MGLS, and IEEE.

H. Rutter, Geophysical Exploration Consultants Pty Ltd, Suite 106, 104 Mount Street, Heidelberg, Vic. 3084.



Don Esdale holds a BSc from the University of Queensland and a Graduate Diploma in Applied Geophysics from the University of New South Wales. He joined Western Mining Corporation in 1965, and spent the next ten years in Kalgoorlie actively involved in exploration for base metals, gold and uranium. After a transfer to Adelaide and a brief period with Western Mining's Petroleum section, he was appointed to his present position of Chief Minerals Geophysicist. He is a member of the ASEG, SEG, EAEG, and PESA.

D. J. Esdale, Western Mining Corporation Ltd, PO Box 409, Unley, SA 5061.

Seismic demarcates shale-out line

O. D. Singh and M. K. Tripathi

Introduction

Initial seismic activity in South Cambay basin of Gujarat, India resulted in the discovery of Anklesvar and Kosamba oil fields. Some of the lower layers of the producing sands of Anklesvar field were found to be absent in the Kosamba field. The theme of this paper is to highlight the success met in demarcation of shale-out line for the lowermost Middle Eocene sands by seismic studies. The main text is divided into three parts. The earlier picture describes the exploration results based on conventional seismic data, while the recent picture is the outcome of digitally acquired CDP data. The additional studies provide further evidence of the existence of the shale-out line in question.

Earlier picture

Two structural highs namely Anklesvar and Kosamba were mapped in the early 1960s on the basis of conventional seismic work. Both these proved to be commercial hydrocarbon pools on drilling. Oil production in the northern pool (Anklesvar) is obtained from Middle and Upper Eocene sands from different layers which are grouped as middle and upper sands separated by a thick shale (Raju & Rao 1975). In the southern pool (Kosamba), oil is produced from Upper Eocene and top layer of Middle Eocene sands.

The drilling results (Wells 1 & 2) on these two structures (Fig. 1) opened a play for the southern limits of Middle Eocene

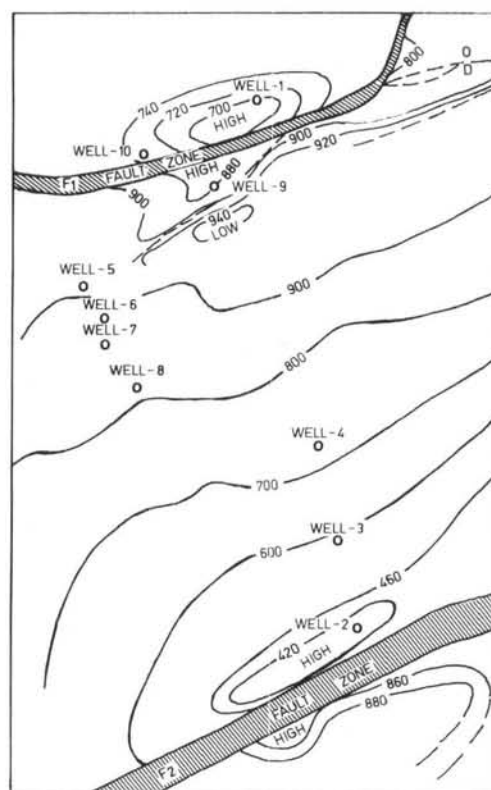


Fig 1 Structural contour trend (in ms) shown at Oligocene level in the area of study.